

1-

```
postgres=# create table students (id int PRIMARY KEY , name varchar(50) ,email varchar(50), address varchar(50) , track_id int ,FOREIGN KEY (track_id) REFERENCES track(id));
CREATE TABLE
postgres=#
```

```
postgres=# create table subject (id int PRIMARY KEY,sub_name varchar(50), max_score int );
CREATE TABLE
```

```
postgres=# create table track ( id int PRIMARY KEY,
postgres=# name text );
CREATE TABLE
```

```
postgres=# create table exam (id int PRIMARY KEY , date Date);
CREATE TABLE
postgres=#
```

```
postgres=# create table grades (grade int , stu_id int , sub_id int , exam_id int ,PRIMARY KEY (stu_id, sub_id, exam_id) , FOREIGN KEY (stu_id) REFERENCES students(id),FOREIGN KEY (sub_id) REFERENCES subject(id),FOREIGN KEY (exam_id) REFERENCES exam(id));
CREATE TABLE
postgres=#
```

```
postgres=# create table stu_sub ( stu_id int , sub_id int , PRIMARY KEY (stu_id, sub_id ), FOREIGN KEY (stu_id) REFERENCES students(id) , FOREIGN KEY (sub_id) REFERENCES subject(id));
CREATE TABLE
postgres=#
```

2-

```
postgres=# INSERT INTO track (id, name) VALUES
(1,'python'),
(2,'python'),
(3,'Mechanical');
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO students (id,name, email, address, track_id) VALUES
(1,'Menna Mamdouh','menna@gmail.com','nemis',1) ,(2,'arwa ahmed','arwa@gmail.com','nemis',2), (3,'aya emad','aya@gmail.com','nemis',3);
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO subject (id,sub_name, max_score) VALUES
(1,'Database Systems', 95),
(2,'Circuit Theory', 90),
(3,'Algorithms', 92);
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO exam (id,date) VALUES
(1,'2023-10-15'),
(2,'2023-11-20'),
(3,'2023-12-05');
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO stu_sub (stu_id, sub_id) VALUES
(1, 1),
(3, 3),
(2, 2);
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO grades (stu_id, sub_id, exam_id, grade) VALUES
(1, 1, 1, 88),
(3, 3, 3, 92),
(2, 2, 2, 90);
INSERT 0 3
postgres=#
```

```
postgres=# INSERT INTO track_sub (track_id, sub_id) VALUES
(1, 1),
(2, 2),
(3, 3);
INSERT 0 3
postgres=#
```

3-

```
postgres=# ALTER TABLE students ADD COLUMN birth_date DATE;
ALTER TABLE
postgres=#
```

4-

```
postgres=# CREATE TYPE gender_enum AS ENUM ('Male', 'Female');
CREATE TYPE
```

```
postgres=# SELECT COUNT(gender) AS count, gender FROM students GROUP BY gender;
```

5-Foreign key are already included in the table creation.

7-

```
postgres=# SELECT name
FROM students WHERE name LIKE 'A%';
name
-----
(0 rows)

postgres=# SELECT name
FROM students WHERE name LIKE 'A';
name
-----
(0 rows)
```

8-

```
postgres=# SELECT sub_name, max_score
FROM subject
ORDER BY max_score DESC;
sub_name | max_score
-----+-----
Database Systems | 95
Algorithms | 92
Circuit Theory | 90
(3 rows)

postgres=#
```

9-

```
postgres=# SELECT sub_name, max_score
FROM subject
ORDER BY max_score DESC
LIMIT 1;
sub_name | max_score
-----+-----
Database Systems | 95
(1 row)
```

Lab 2

1-

```
postgres=# SELECT COUNT (name) FROM students
WHERE name = 'arwa';
 count
-----
      0
(1 row)
```

2-

```
postgres=# SELECT COUNT(gender) AS count, gender FROM students GROUP BY gender;
 count | gender
-----+-----
      0 |
(1 row)
```

3-

```
postgres=# SELECT
COUNT(name) AS name_count
FROM students HAVING COUNT(name) > 2;
 name_count
-----
          3
```

4-

```
postgres=# SELECT students.name AS student_name, track.name AS track_name
FROM students JOIN track ON students.track_id = track.id;
 student_name | track_name
-----+-----
 Menna Mamdouh | python
 arwa ahmed    | python
 aya emad      | Mechanical
(3 rows)
```

5-

```
postgres=# SELECT* FROM students
WHERE track_id NOT IN (SELECT id FROM track WHERE name = 'python');
 id | name | email | address | track_id | birth_date
-----+-----+-----+-----+-----+-----
  3 | aya emad | aya@gmail.com | nemis | 3 |
(1 row)
```

```
postgres=#
```